EXPERIMENT-6
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Objective :-WAP to train and evaluate a Recurrent Neural Network using PyTorch Library to predict the next value in a sample time series dataset.

Description :-

1.Dataset

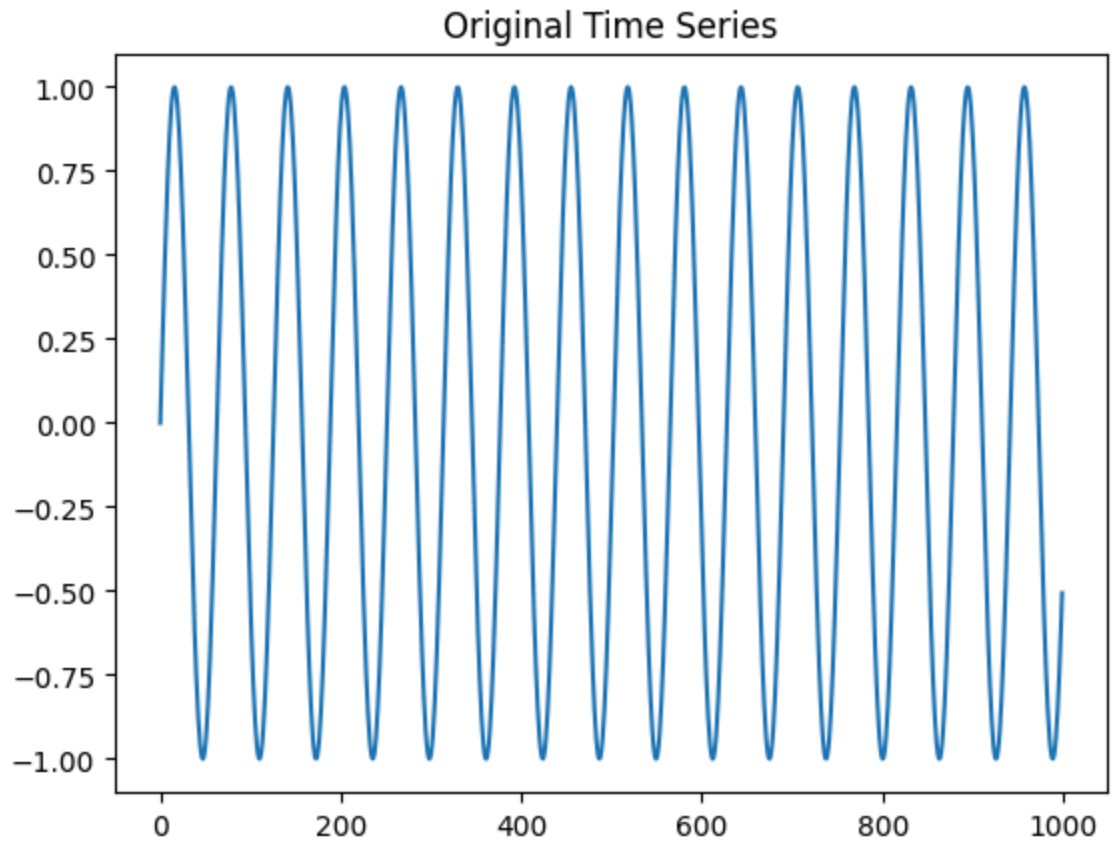
- Generated sine wave
- Converted into sequences for supervised learning

2.Model

- `nn.RNN` used for sequence learning
- Hidden state stores past info
- Fully connected layer gives final prediction

Source Code :-

```
import torch
import torch.nn as nn
import numpy as np
import matplotlib.pyplot as plt

# Generate sine wave data
time_steps = np.linspace(0, 100, 1000)
data = np.sin(time_steps)

# Plot data
plt.plot(data)
plt.title("Original Time Series")
plt.show()
```




```
def create_sequences(data, seq_length):
    sequences = []
    labels = []

    for i in range(len(data) - seq_length):
        seq = data[i:i+seq_length]
        label = data[i+seq_length]
        sequences.append(seq)
        labels.append(label)

    return np.array(sequences), np.array(labels)

seq_length = 20
X, y = create_sequences(data, seq_length)

# Convert to tensors
X = torch.tensor(X, dtype=torch.float32).unsqueeze(-1)
```

```
y = torch.tensor(y, dtype=torch.float32)
split = int(0.8 * len(X))

X_train, X_test = X[:split], X[split:]
y_train, y_test = y[:split], y[split:]
class RNNModel(nn.Module):
    def __init__(self, input_size=1, hidden_size=32, num_layers=1):
        super(RNNModel, self).__init__()

        self.rnn = nn.RNN(input_size, hidden_size, num_layers,
batch_first=True)
        self.fc = nn.Linear(hidden_size, 1)

    def forward(self, x):
        out, _ = self.rnn(x)   # out: (batch, seq_len, hidden)
        out = out[:, -1, :]    # take last time step
        out = self.fc(out)
        return out

model = RNNModel()
criterion = nn.MSELoss()
optimizer = torch.optim.Adam(model.parameters(), lr=0.001)
epochs = 40
train_losses = []

for epoch in range(epochs):
    model.train()

    output = model(X_train)
    loss = criterion(output.squeeze(), y_train)

    optimizer.zero_grad()
    loss.backward()
    optimizer.step()

    train_losses.append(loss.item())
    if epoch%5 == 0 :
     print(f"Epoch [{epoch+1}/{epochs}], Loss: {loss.item():.6f}")
```

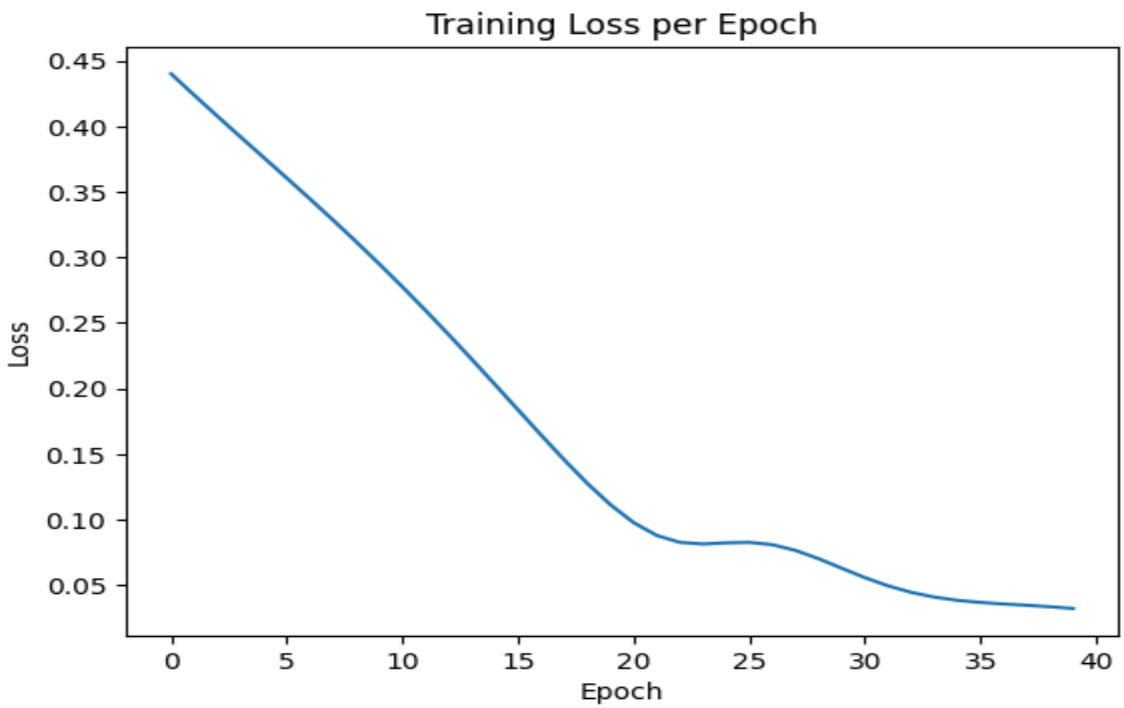
OUTPUT :-

```
Epoch [1/40], Loss: 0.439948
Epoch [6/40], Loss: 0.360510
Epoch [11/40], Loss: 0.277531
Epoch [16/40], Loss: 0.183826
Epoch [21/40], Loss: 0.097799
Epoch [26/40], Loss: 0.082736
Epoch [31/40], Loss: 0.055853
Epoch [36/40], Loss: 0.036909
```

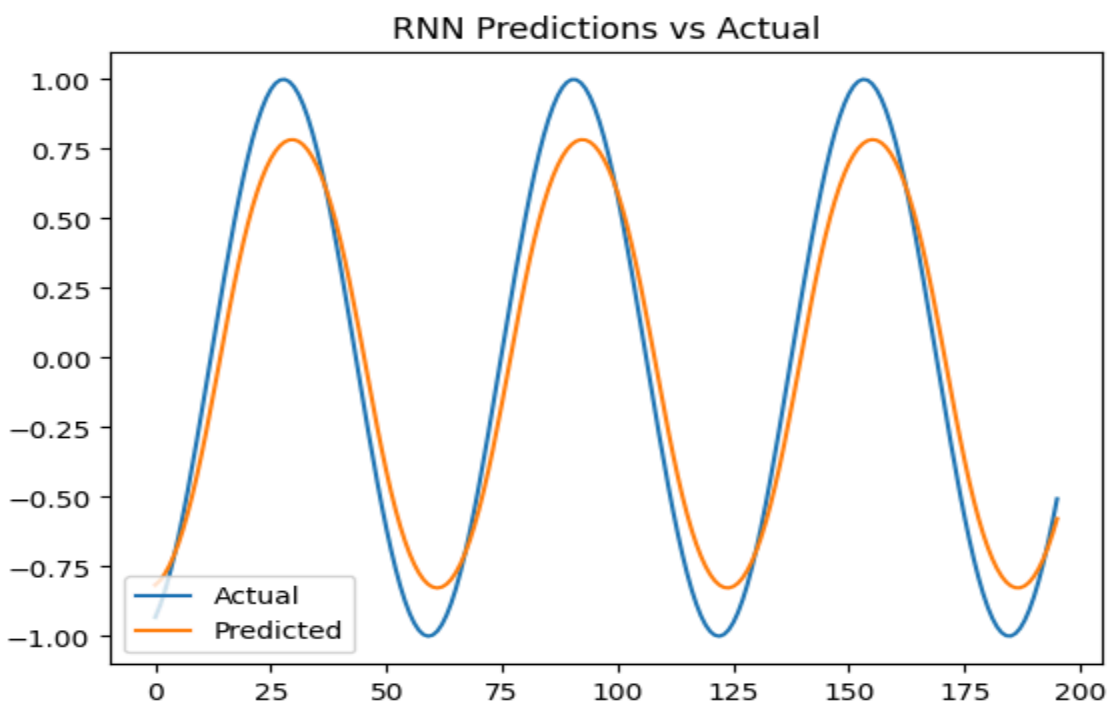
```
model.eval()
with torch.no_grad():
    predictions = model(X_test).squeeze().numpy()
    actual = y_test.numpy()
from sklearn.metrics import mean_squared_error

mse = mean_squared_error(actual, predictions)
rmse = np.sqrt(mse)

print(f"MSE: {mse:.6f}")
print(f"RMSE: {rmse:.6f}")
```

```
OUTPUT :-
MSE: 0.029779
RMSE: 0.172566
```

```
plt.plot(train_losses)
plt.title("Training Loss per Epoch")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.show()
```




```
plt.plot(actual, label="Actual")
plt.plot(predictions, label="Predicted")
plt.legend()
plt.title("RNN Predictions vs Actual")
plt.show()
```



MY COMMENTS :-

From this experiment, Successfully implemented a Recurrent Neural Network (RNN) using PyTorch for time series prediction.

Learning from Experiment :-

- RNN processes sequential data using hidden states
- Previous inputs influence future predictions
- Suitable for time series forecasting
- Predictions closely follow sine wave pattern
- RNN captures temporal dependencies